REAL ESTATE ANALYTICS PROJECT

Leveraging SQL & Power BI for Strategic Property Insights

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"Without data, you're just another person with an opinion."
W. Edwards Deming

Introduction

In an evolving world with a growing demand for real estate, it has become increasingly common to face urgent situations when buying or renting properties. In order to better understand this dynamic activity and provide meaningful solutions, I chose to develop a project based on the **"Comprehensive Real Estate Agency Dataset."**

Objective

This project aims to gain a comprehensive view of the real estate sector and its trends by analyzing data from an agency to deliver **actionable insights** that support **strategic decision-making**. The study focuses on **sales performance, property visits, and market trends**, while highlighting skills in **data modeling, Power BI** and **SQL** to extract and interpret key information effectively.

Description of the main Tables

The dataset powering this analysis is structured around five interconnected tables, each offering key insights into the real estate ecosystem. These tables : **Agents**, **Customers**, **Properties**, **Sales**, and **Visits** are collectively capture the full scope of operational activity, from client interactions to transaction outcomes. Below is an overview of each table along with its relevant variables.

Agents Table

Key variables :

- AgentID
- FirstName
- LastName
- Email
- Phone

Customers Table

Key variables :

- CustomerID
- FirstName
- LastName
- Email
- Phone

Properties Table

Key variables :

- PropertyID
- Location
- PropertyType
- Size_sqm
- PriceUSD

Sales Table

Key variables :

- SaleID
- PropertyID
- CustomerID
- AgentID
- SaleID
- SalePrice

Visits Table

Key variables :

- VisitID
- PropertyID
- CustomerID
- AgentID
- VisitDate

Database Creation

To build our database infrastructure, we utilized two essential SQL tools: **SQL Server Express 2021** for database deployment and **SQL Server Management Studio 2021** for efficient data management and query execution. Below are our different tables :

```sql
-- Create the database
CREATE DATABASE RealEstateAgencyDB;

-- Switch to the new database
USE RealEstateAgencyDB;

-- Supprimer la table si elle existe
IF OBJECT_ID('dbo.Agents', 'U') IS NOT NULL
    DROP TABLE dbo.Agents;
-- AGENTS TABLE
CREATE TABLE Agents (
    AgentID INT PRIMARY KEY,
    FirstName NVARCHAR(50) NOT NULL,
    LastName NVARCHAR(50) NOT NULL,
    Phone NVARCHAR(50),
    Email NVARCHAR(100) UNIQUE
);
```

```sql
IF OBJECT_ID('dbo.Customers', 'U') IS NOT NULL
    DROP TABLE dbo.Customers;

-- CUSTOMERS TABLE
CREATE TABLE Customers (
    CustomerID INT PRIMARY KEY,
    FirstName NVARCHAR(50) NOT NULL,
    LastName NVARCHAR(50) NOT NULL,
    Phone NVARCHAR(50),
    Email NVARCHAR(100)

);

-- Supprimer la table si elle existe
IF OBJECT_ID('dbo.Properties', 'U') IS NOT NULL
    DROP TABLE dbo.Properties;

-- PROPERTIES TABLE
CREATE TABLE Properties (
    PropertyID INT PRIMARY KEY,
    PropertyType NVARCHAR(50), -- Apartment, House, Office, etc.
    Location NVARCHAR(50),
    Size_sqm DECIMAL(10,2),
    PriceUSD DECIMAL(15,2)

);
```

```sql
-- SALES TABLE
CREATE TABLE Sales (
    SaleID INT PRIMARY KEY,
    PropertyID INT NOT NULL,
    CustomerID INT NOT NULL,
    AgentID INT NOT NULL,
    SaleDate DATE,
    SalePrice DECIMAL(15,2),
    CONSTRAINT FK_Sales_Properties FOREIGN KEY (PropertyID)
        REFERENCES Properties(PropertyID)
        ON UPDATE CASCADE
        ON DELETE NO ACTION,
    CONSTRAINT FK_Sales_Customers FOREIGN KEY (CustomerID)
        REFERENCES Customers(CustomerID)
        ON UPDATE CASCADE
        ON DELETE NO ACTION,
    CONSTRAINT FK_Sales_Agents FOREIGN KEY (AgentID)
        REFERENCES Agents(AgentID)
        ON UPDATE CASCADE
        ON DELETE NO ACTION
);
```

```sql
-- VISITS TABLE
CREATE TABLE Visits (
    VisitID INT PRIMARY KEY,
    CustomerID INT NOT NULL,
    PropertyID INT NOT NULL,
    AgentID INT NOT NULL,
    VisitDate DATE,
    CONSTRAINT FK_Visits_Properties FOREIGN KEY (PropertyID)
        REFERENCES Properties(PropertyID)
        ON UPDATE CASCADE
        ON DELETE NO ACTION,
    CONSTRAINT FK_Visits_Customers FOREIGN KEY (CustomerID)
        REFERENCES Customers(CustomerID)
        ON UPDATE CASCADE
        ON DELETE NO ACTION,
    CONSTRAINT FK_Visits_Agents FOREIGN KEY (AgentID)
        REFERENCES Agents(AgentID)
        ON UPDATE CASCADE
        ON DELETE NO ACTION
);
```

Uploading CSV files to the database

```sql
--Providing Data To Our Tables Using Our Csv files.
INSERT INTO Agents (AgentID,FirstName, LastName, Phone, Email)
SELECT AgentID,FirstName, LastName, Phone, Email FROM Agents_temp;

INSERT INTO Customers (CustomerID,FirstName,LastName,Phone,Email)
SELECT CustomerID,FirstName,LastName,Phone,Email FROM Customers_temp;

INSERT INTO Properties (PropertyID,PropertyType, Location, Size_sqm, PriceUSD)
SELECT PropertyID,PropertyType, Location, Size_sqm, PriceUSD FROM Properties_temp;

INSERT INTO Sales (SaleID,PropertyID, CustomerID,AgentID, SaleDate, SalePrice)
SELECT SaleID,PropertyID, CustomerID,AgentID, SaleDate, SalePrice FROM Sales_temp;

INSERT INTO Visits (VisitID,PropertyID, CustomerID, AgentID, VisitDate)
SELECT VisitID,PropertyID, CustomerID, AgentID, VisitDate FROM Visits_temp;

--We drop unnecessary tables
DROP TABLE IF EXISTS Agents_temp, Customers_temp, Properties_temp, Sales_temp, Visits_temp;
```

Some Business Questions

```sql
--Q1 : Total number of Properties
SELECT COUNT(*) AS total_properties
FROM Properties;
```

	total_properties
1	1000

```sql
--Q2 : Total number of Properties by Type
SELECT PropertyType, COUNT(*) AS total_properties
FROM Properties
GROUP BY PropertyType
ORDER BY total_properties DESC;
```

	PropertyType	total_properties
1	Apartment	215
2	Villa	208
3	Retail	195
4	Warehouse	192
5	Office	190

```sql
--Q6 : Number of Sales made by each Agent
SELECT a.AgentID, a.FirstName, a.LastName, COUNT(s.SaleID) AS Sales_count
FROM Sales s
JOIN Agents a ON s.AgentID = a.AgentID
GROUP BY a.AgentID, a.FirstName, a.LastName
ORDER BY sales_count DESC;
```

	AgentID	FirstName	LastName	Sales_count
1	91	Francisco	Williams	29
2	31	Gordon	Wilson	28
3	87	Michelle	Davis	28
4	93	Samantha	Vargas	28
5	17	Kelli	Davis	27
6	36	Taylor	Clark	26
7	6	Larry	Holloway	26
8	46	Cody	Ramsey	26

```sql
--Q7 : Turnover per agent to provide the sales and profit generated by each agent
SELECT
    a.AgentID,
    a.FirstName,
    a.LastName,
    ROUND(SUM(s.SalePrice), 2) AS Total_sales,
    ROUND(SUM(CASE
                WHEN s.SalePrice >= p.PriceUSD
                THEN s.SalePrice - p.PriceUSD
                ELSE 0
              END), 2) AS Total_profit
FROM Sales s
JOIN Agents a ON s.AgentID = a.AgentID
JOIN Properties p ON s.PropertyID = p.PropertyID
GROUP BY a.AgentID, a.FirstName, a.LastName
ORDER BY Total_sales DESC;
```

	AgentID	FirstName	LastName	Total_sales	Total_profit
1	93	Samantha	Vargas	24215411.00	10474653.00
2	91	Francisco	Williams	24176580.00	12726869.00
3	31	Gordon	Wilson	23283861.00	13209631.00
4	21	Robert	Miller	21440733.00	13697740.00
5	40	Mark	Lopez	20987705.00	12311253.00
6	14	Laura	Martin	20850801.00	11116662.00
7	87	Michelle	Davis	20640884.00	9298269.00
8	46	Cody	Ramsey	20495310.00	8362805.00
9	61	Tammy	Walker	20447661.00	11157920.00
10	20	Tammy	Cohen	20395399.00	9345972.00
11	73	Nathan	Stevens	20118590.00	9868233.00

```sql
--Q8 : Agent performance (sales/visit conversion rate)
SELECT
    a.AgentID,
    a.FirstName,
    a.LastName,
    COUNT(DISTINCT v.VisitID) AS Total_visits,
    COUNT(DISTINCT s.SaleID) AS Total_sales,
    ROUND(COUNT(DISTINCT s.SaleID) * 100.0 / COUNT(DISTINCT v.VisitID), 2) AS Conversion_rate
FROM Agents a
LEFT JOIN Visits v ON a.AgentID = v.AgentID
LEFT JOIN Sales s ON a.AgentID = s.AgentID
GROUP BY a.AgentID, a.FirstName, a.LastName
HAVING COUNT(DISTINCT v.VisitID) > 0
ORDER BY Conversion_rate DESC;
```

	AgentID	FirstName	LastName	Total_visits	Total_sales	Conversion_rate
1	91	Francisco	Williams	38	29	76.320000000000
2	100	Tyler	Maddox	36	24	66.670000000000
3	73	Nathan	Stevens	36	23	63.890000000000
4	17	Kelli	Davis	44	27	61.360000000000
5	67	Joshua	Olson	40	24	60.000000000000
6	31	Gordon	Wilson	48	28	58.330000000000
7	54	Sharon	Ross	40	23	57.500000000000
8	6	Larry	Holloway	47	26	55.320000000000
9	36	Taylor	Clark	49	26	53.060000000000

```sql
--Q17 : Top 5 months with the most sales
SELECT TOP 5 YEAR(SaleDate) AS Year, MONTH(SaleDate) AS Month,
       COUNT(SaleID) AS Sales_Count
FROM Sales
GROUP BY YEAR(SaleDate), MONTH(SaleDate)
ORDER BY Sales_Count DESC;
```

	Year	Month	Sales_Count
1	2024	12	102
2	2024	3	101
3	2024	5	100
4	2024	7	94
5	2023	11	93

```sql
--Q18 : Agents with the most visits
SELECT a.AgentID, a.FirstName, a.LastName,
       COUNT(v.VisitID) AS Visit_Count
FROM Agents a
JOIN Visits v ON a.AgentID = v.AgentID
GROUP BY a.AgentID, a.FirstName, a.LastName
ORDER BY Visit_Count DESC;
```

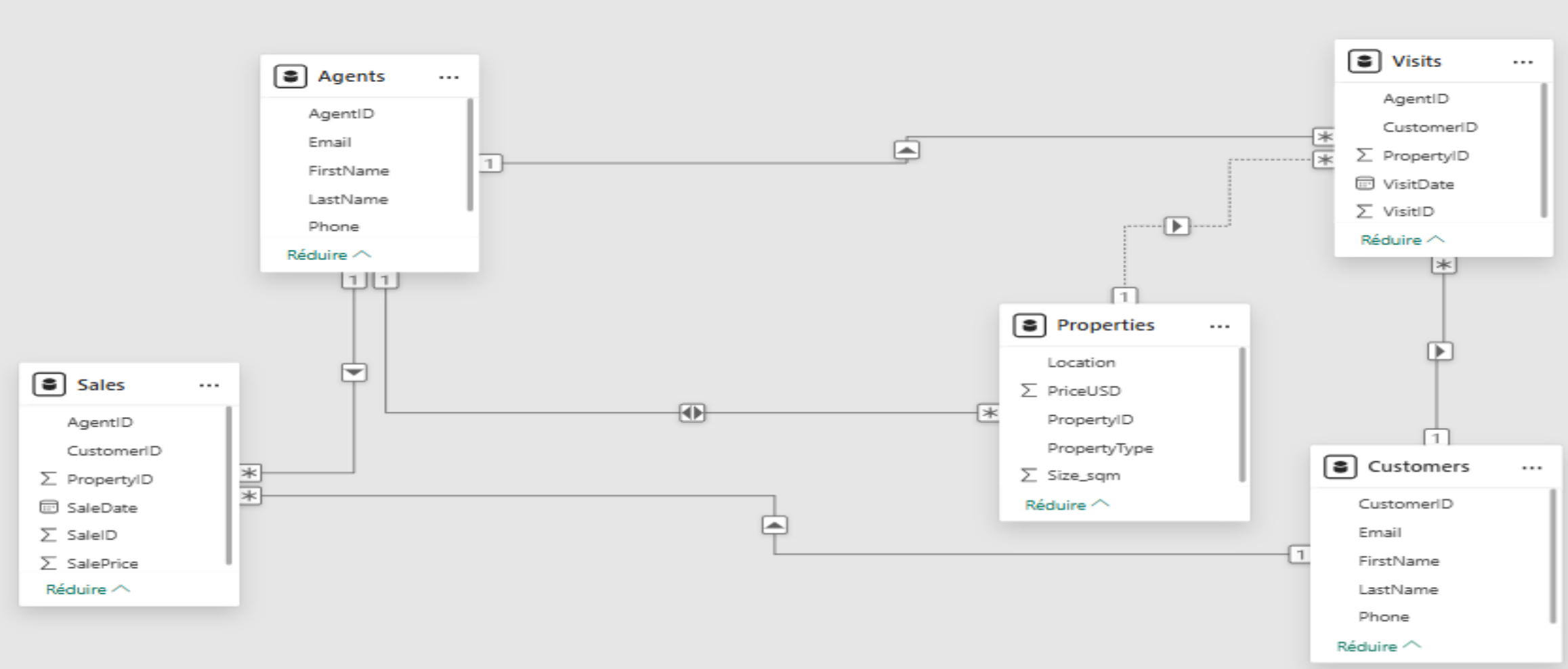
	AgentID	FirstName	LastName	Visit_Count
1	23	Carolyn	Gibbs	66
2	15	Julie	Barrett	66
3	93	Samantha	Vargas	64
4	40	Mark	Lopez	64
5	92	Carolyn	Terry	63
6	35	Ryan	Gould	60
7	98	Jeremy	Wade	60
8	24	Nicholas	Shaffer	60
9	33	Paul	Brooks	60
10	80	Kim	Jackson	60
11	57	Brian	Watson	59
12	45	Joshua	Macias	59
13	26	Edward	Rush	59

```sql
--Q20 : Percentage of properties sold within 30 days
SELECT
    ROUND(
        SUM(CASE
                WHEN DATEDIFF(DAY, v.VisitDate, s.SaleDate) <= 30 THEN 1
                ELSE 0
            END) * 100.0 / COUNT(*), 2
    ) AS sold_within_30_days
FROM Sales s
JOIN Properties p ON s.PropertyID = p.PropertyID
JOIN Visits v ON s.PropertyID = v.PropertyID
WHERE v.VisitDate <= s.SaleDate;
```

	sold_within_30_days
1	17.16000000000

Power BI Analysis

Data Model



The dashboard is on its way.
I'll be sharing it shortly.
Please feel free to share any feedback or suggestions you may have.
Thanks and see you soon!